Fabricio Giusti Oliveira Monteiro

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Dual Citizenship - Brazil and Italy

EDUCATION

Purdue University, College of Engineering, West Lafayette, IN *August 2024 – May 2028*
Bachelor of Science in Mechanical Engineering *GPA: 3.91/4.0*
Minor in Global Engineering Studies (GEARE Program)

RELEVANT EXPERIENCE

Advanced Computation and Artificial Intelligence (ACAI) Lab *January 2026 – Present*
Undergraduate Research Assistant, West Lafayette, IN

- Implemented autonomous manipulation algorithms on the Mobile ALOHA platform, focusing on physical intelligence in unstructured environments.
- Fine-tuned Vision-Language-Action (VLA) models and LLMs in Python to successfully automate bookshelf organization tasks.
- Developed MuJoCo simulations to train and validate manipulation policies prior to real-world deployment.

Engineering Projects in Community Service (EPICS) – Mobility (MOBI)
Design Lead, West Lafayette, IN *August 2024 – May 2025*

- Directed a 5-member team in partnership with USA Blind Hockey to design and 3D-print puck prototypes capable of withstanding 500 N of force, testing different materials, closures (snap-fit and screwed), and infill densities.
- Conducted real-user testing with a hockey player, validating durability while improving sound output and waterproofing for accessibility.

Nostradamus Project
Creator & Developer, West Lafayette, IN *September 2024 – December 2024*

- Engineered a Python program that estimates hurricane probability by month for any city worldwide using the National Centers for Environmental Information historical data (1840–present).
- Enforced geocoding, data processing (pandas), and an interactive Folium heatmap with a location marker and risk tooltip, outputting a personalized risk percentage and navigable map.

LEADERSHIP & COMMUNITY INVOLVEMENT

Boiler Soccer Robot Association (BSRA) *November 2025 – Present*
Co-Founder & President, West Lafayette, IN

- Founded humanoid robotics club at Purdue to compete in the RoboCup Humanoid League, an international competition for fully autonomous multi-robot soccer teams.
- Led recruitment and secured necessary funding to launch the team, establishing clear development goals for the mechanical, electrical, and software sub teams.

BRASA – Brazilian Student Association at Purdue University *September 2024 - Present*
Workshop Director, West Lafayette, IN

- Designed and delivered workshops for 60+ attendees (50% growth from prior year) in a 200+ member community, creating tailored presentations to student career needs.
- Directed a 4-member analyst team to research audience interests to shape workshop topics and boost engagement.

AWARDS & CERTIFICATES

- H. William Bottomley Research Scholarship, Purdue University *January 2025*
- Dean's List and Semester Honors Recognition, Purdue University *August 2024 – December 2025*
- Machine Learning Specialization, Stanford University *August 2025*

SKILLS

Technical: Fusion 360, Siemens NX, Teamcenter, Catia, 3Dexperience, MuJoCo.
Programming Languages: Python, MATLAB, C.
Languages: Portuguese (Native), English (Fluent), Spanish (Fluent), Italian (Basic), Mandarin (Basic)
Interests: Soccer, Traveling, Machine learning, Guitar, Automation.